

Dilip Sahu

B.Tech Computer Science & Engineering | NIT Mizoram
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EDUCATION

National Institute of Technology Mizoram

B.Tech in Computer Science & Engineering | CGPA: **7.98/10**

Aizawl, Mizoram

2024 – 2028

Coursework: Artificial Intelligence, Machine Learning, C Programming, Data Structures & Algorithms, OOP, Computer Organization & Architecture, DBMS, Operating Systems, Computer Networks, Distributed Systems, Cloud Computing, Probability & Stochastics

Impulse Academy

Class XII – RBSE | **86%**

Jaipur, Rajasthan

2023

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Java, JavaScript, SQL

Frameworks & Databases: FastAPI, Node.js, React, React Native, Redis, PostgreSQL, MySQL, MongoDB

AI/ML: Neural Networks, Deep Learning, NLP, **Multi-stage Transfer Learning** (layer freezing, discriminative fine-tuning, variable learning rates), Computer Vision (OpenCV, YOLO), **PyTorch**, **TensorFlow**

Algorithms: Data Structures & Algorithms, Graph Algorithms, Cryptography, Competitive Programming

Tools: Git, GitHub, Linux, VS Code | Quick learner of new technologies

EXPERIENCE

Zetheta Algorithms Pvt. Ltd.

Software Engineer Intern (Project-Based)

Remote

Jul 2026 – Aug 2026

- **FastAPI reconciliation engine** – 100,000+ multi-bank transactions; **RapidFuzz**-based entity matching
- **Automated data-ingestion pipelines** – 3+ banking formats; **Pydantic** schema validation
- **Tamper-evident PostgreSQL audit storage** – **SHA-256** integrity; 100% transaction traceability

PROJECTS

CyberGuard-AI – Real-Time Scam Detection [GitHub](#)

Edge AI & NLP

- Built a **multi-stage Transfer Learning** model for scam-intent detection: frozen layers first, then **discriminative fine-tuning** with layer-wise learning rates (avoids **catastrophic forgetting**)
- **O(1) Bloom Filter** for instant known-caller and spam lookup
- **Temporal sliding window** keeps conversation context for live scam analysis

Coding & AI Club NIT Mizoram Website [Website](#) | [GitHub](#)

Full-Stack Engineering

- Cut form-submission time from **2000ms to <100ms (95%)** using hover-intent prefetching via **DOM tree traversal**
- **Hash Set**-based duplicate check across all members of multi-user team registrations in **O(N)** time
- **Exponential backoff + jitter** for fault-tolerant API sync

IoT-Based Bus-Tracking System [Website](#) | [GitHub](#) | [Images](#)

Real-Time Data & Intelligent Systems

- **Node.js** backend designed for **10,000+ req/sec** (5,000+ peak); **Redis** FIFO queue + Server-Sent Events for live updates
- **Haversine**-based ETA with **<50ms** map updates (Leaflet, PostgreSQL)

TourAI – Predictive AI Travel Planner [GitHub](#)

Machine Learning

- **Prophet** forecasting of tourism demand on a 15-year Mizoram dataset
- **Random Forest** model to judge destination and itinerary feasibility
- **FastAPI** backend serving predictions for travel planning

ACHIEVEMENTS & LEADERSHIP

- **5G Innovation Hackathon 2026:** Top 50 National Finalist; CyberGuard-AI – INR 1,00,000 seed funding
- **Hackathons:** 1st Place – 3+ institute-level AI/innovation hackathons; 1st Runner-Up – Internal SIH
- **Amazon ML Summer Challenge 2026:** 95.29/100
- **Kaggle:** Global Rank 1,289 – Deep Past Language Challenge 2026
- **Competitive Programming:** LeetCode + Codeforces + Striver's Sheet – **550+ problems**
- **Selective Programs:** Polaris Fellowship; HPCL-Mittal Energy Quest 3.0 (Pre-final stages)
- **Leadership:** Vice President – Coding & AI Club; Website Coordinator – T&P Office, NIT Mizoram
- **Certificates:** Fundamentals of Cybersecurity (Zscaler); Drone Workshop Boot-Camp (Jet Aerospace, IIT Delhi & NIT Silchar); React Native Developer (HackerRank)